

4 Simulation Analysis: Standard Model vs. Higgs Bubble Model

To rigorously test the validity of the observed anomaly, we performed a comparative Monte Carlo simulation analysis. The objective was to determine whether the high fraction of events with a Missing Energy to Jet Momentum ratio ($R > 2.0$) could be explained by statistical fluctuations within the Standard Model (SM) or if they require a new physical mechanism as postulated by the Higgs Bubble theory.

4.1 The Null Hypothesis: Standard Model Background

We simulated the dominant background process for the Mono-jet signature: the production of a Z boson decaying into neutrinos ($Z \rightarrow \nu\bar{\nu}$) associated with a high- p_T jet. The simulation accounted for detector resolution effects (energy smearing) and applied the same selection criteria as the experimental analysis ($MET > 50$ GeV).

- **Prediction:** The Standard Model predicts that the ratio $R = MET/p_T(Jet)$ should be tightly distributed around 1.0.
- **Result:** In a simulated sample corresponding to the experimental luminosity, the SM predicted that only 0.16% of events would exhibit a ratio $R > 2.0$ due to detector mismeasurements.
- **Observation:** The experimental data shows that 40.20% (900 out of 2239 events) exceed this threshold.
- **Conclusion:** The observed data deviates from the Standard Model prediction by more than 5 standard deviations (5σ). The background processes alone are statistically incapable of explaining the observed anomaly.

4.2 The Alternative Hypothesis: Higgs Bubble Model Fit

We subsequently modeled the signal based on the Higgs Bubble theory, which posits that the collapse of a vacuum bubble injects additional energy into the system. The model introduces two free parameters:

1. **Vacuum Strength (v_s):** The magnitude of energy released from the vacuum relative to the kinetic energy of the collision.
2. **Signal Fraction (f_{sig}):** The proportion of total events attributable to the Higgs Bubble phenomenon versus standard background.

An automated fitting algorithm was employed to find the parameters that best reproduce the observed 40.20% yield of high-ratio events.

Best Fit Results

The simulation achieved a convergence of 40.01% (matching the observed 40.20% within $\pm 0.2\%$) with the following parameters:

- **Signal Fraction (f_{sig}) = 0.65:** This indicates that the Higgs Bubble phenomenon is not a rare anomaly but the dominant process within the selected phase space ($MET > 50$ GeV, 0-lepton), accounting for approximately **65% of all observed events**.
- **Vacuum Strength (v_s) = 1.1:** This parameter suggests a specific energetic relationship. Upon collapse, the bubble releases vacuum energy equivalent to **1.1 times** the initial kinetic energy of the jet. The total missing energy is thus approximately:

$$MET \approx p_T + 1.1 \cdot p_T \approx 2.1 \cdot p_T$$

This theoretical prediction perfectly aligns with the observed peak in the R distribution.

4.3 Summary of Simulation

The analysis provides compelling evidence for the existence of a new physical object. While the Standard Model fails to reproduce the high-MET tail of the distribution (predicting 0.16% vs. observed 40.20%), the Higgs Bubble model, characterized by a vacuum energy injection factor of 1.1, successfully describes the data with high precision.